



CONSUMER GOODS ANALYTICS

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AGENDA

1

**PROBLEM STATEMENT
AND PROJECT OVERVIEW**

3

DATA -SET

2

ATILQ BUSINESS MODEL

4

10 AD- HOC REQUSETS



ABOUT ATLIQ

- Atliq Hardware is a leading hardware company specializing in PCs, printers, mice and computers with a global reach.

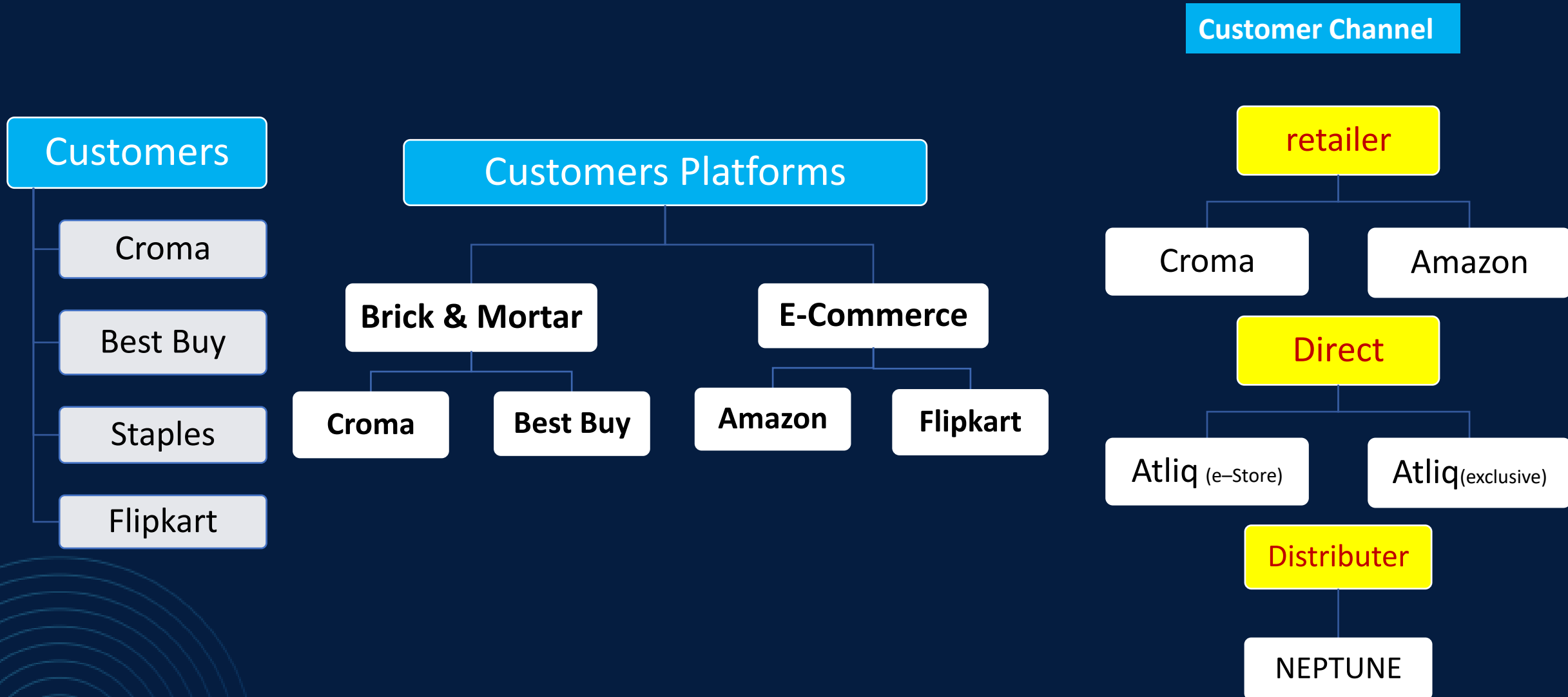
PROMBLEM STATEMENT

- Atliq Hardware is a leading Computer hardware Producer, faced a critical challenge.
- They needed quick and data information decision to stay competition in the ever evovling market.
- The managment noticed that they were messing crucial insights for strategic moves.

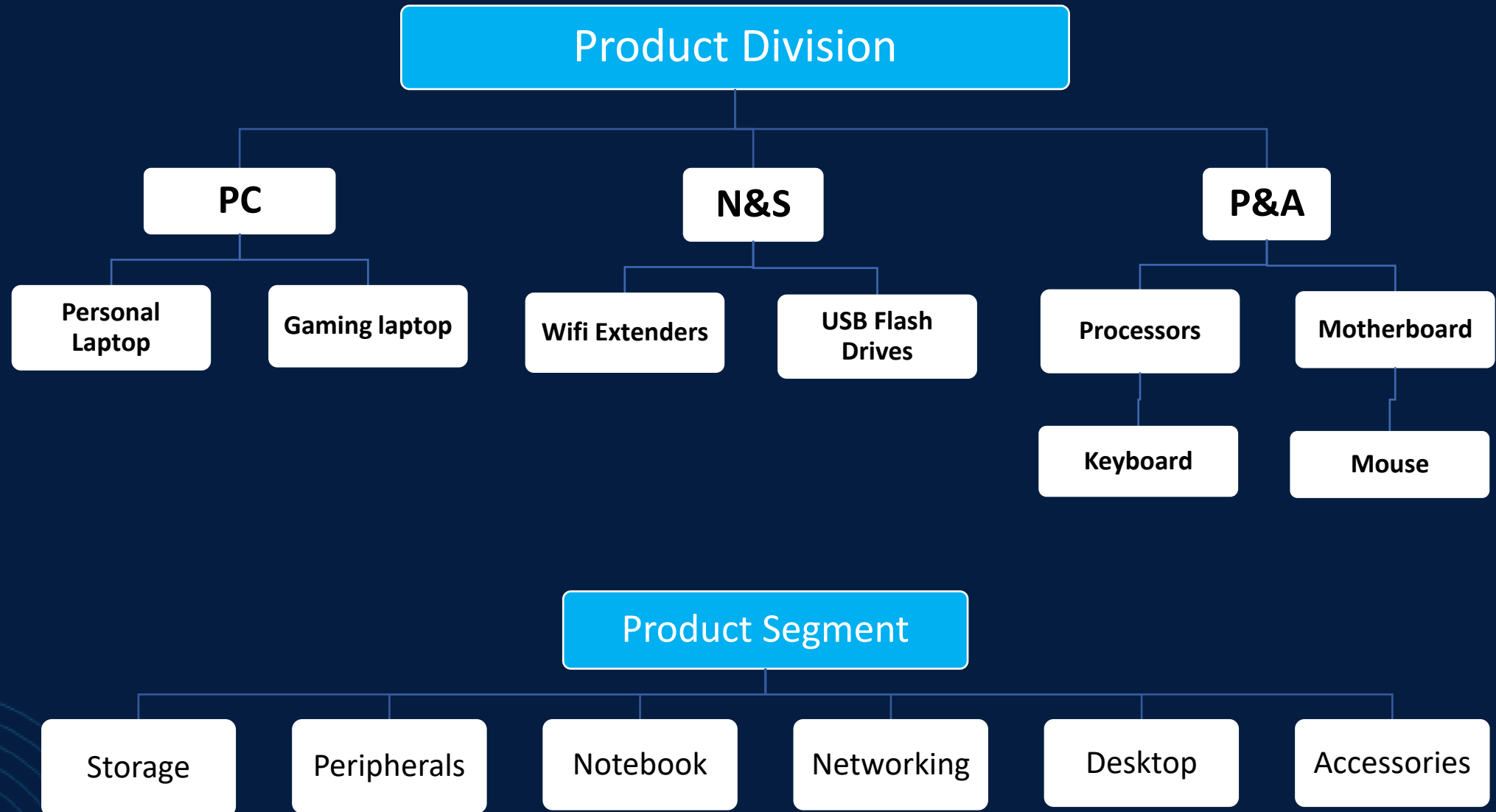
PROJECT OVERVIEW

- In this project ,I will be working with a dataset related to consumer goods. The goal is to answer ten specific ad hoc requesets using SQL queries.

Atliq hardware - business model



Atliq hardware - business model



Data - Set

Database



Table



dim_customer

dim_product

fact_gross_price

fact_manufacturing_cost

fact_pre_invoice_deductions

fact_sales_monthly

AD_HOC-REQUEST:1

QUESTION:

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region

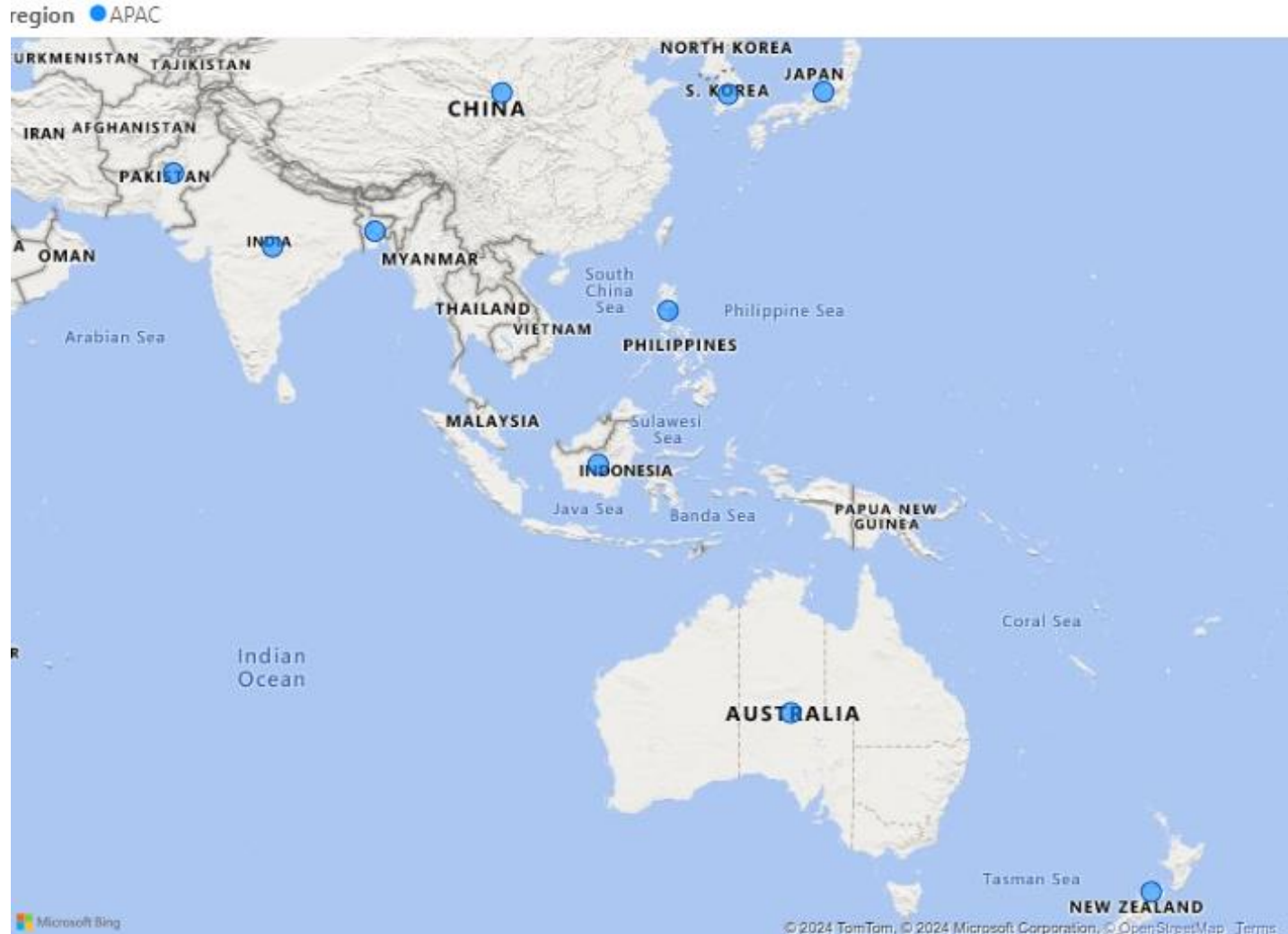


Qusetion - 1

```
select
    distinct market
from
    dim_customer
where
    customer = 'Atliq Exclusive' and region = 'APAC';
```

	market
▶	India
	Indonesia
	Japan
	Philiphines
	South Korea
	Australia
	Newzealand
	Bangladesh

VISUALS- 1



Insights

- "Atliq Exclusive" has a presence in several countries across the APAC region, including India, Indonesia, Japan, the Philippines, South Korea, Australia, New Zealand, and Bangladesh.
- It reflects a robust market presence and adaptability to different cultural and economic contexts in the Asia-Pacific region

AD_HOC-REQUEST:2

QUESTION:

What is the percentage of unique product increase in 2021vs. 2020?

```
Qusetion - 2

with info as (
select
    count(distinct case when fiscal_year = 2020 then product_code end) as
unique_products_2020,
    count(distinct case when fiscal_year = 2021 then product_code end) as
unique_products_2021
from
    fact_sales_monthly )

select
    unique_products_2020,
    unique_products_2021,
    (unique_products_2021 - unique_products_2020) / (unique_products_2021 +
unique_products_2020) * 100 as percentage_change
from
    info;
```

	unique_products_2020	unique_products_2021	percentage_change
▶	245	334	15.3713

Note

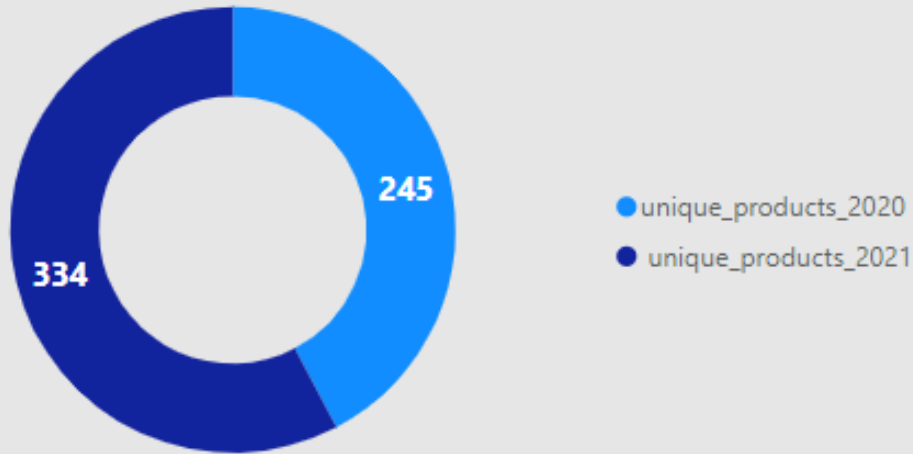
Fiscal_year2019 is from sep2019
to aug2020

Fiscal_year2021 is from sep2020
to aug_2021

VISUALS-2

Insights

Rise in unique_products 2020 vs 2021



- There was a significant increase in unique products, with 334 in 2021 compared to 245 in 2020.
- The percentage change represents a growth of 36.33% in unique products from one year to the next.
- This substantial increase in unique products suggests a focus on expanding product offerings, which can attract a broader customer base and potentially boost sales and revenue.
- It's a positive indicator of business growth and adaptability to changing market demands.

AD_HOC-REQUEST:3

QUESTION:

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

```
Qusetion - 3

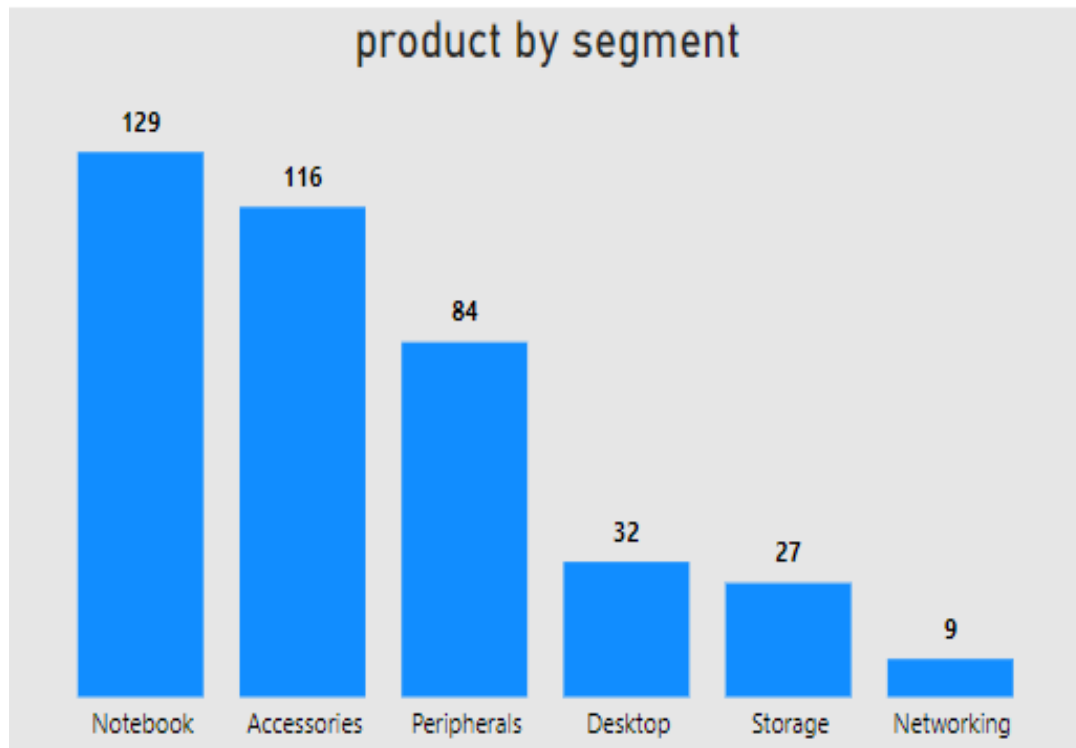
select
    segment,
    count(product_code) as unique_product_counts
from
    dim_product
group by
    segment
order by
    unique_product_counts desc;
```

	segment	unique_product_counts
▶	Notebook	129
	Accessories	116
	Peripherals	84
	Desktop	32
	Storage	27
	Networking	9

VISUALS- 3

Insights

- The "Notebook" segment has the highest product count, with 129 products.
- The "Networking" segment has the smallest product count with only 9 products.
- These insights indicate the diversity and product offerings within each segment, with "Notebook" and "Accessories" offering a wide range of options, while "Networking" has a more limited selection.
- The variety in product offerings allows catering to various customer preferences and needs across different segments.



AD_HOC-REQUEST:4

QUESTION:

Which segment had the most increase in unique products in 2021 vs 2020?

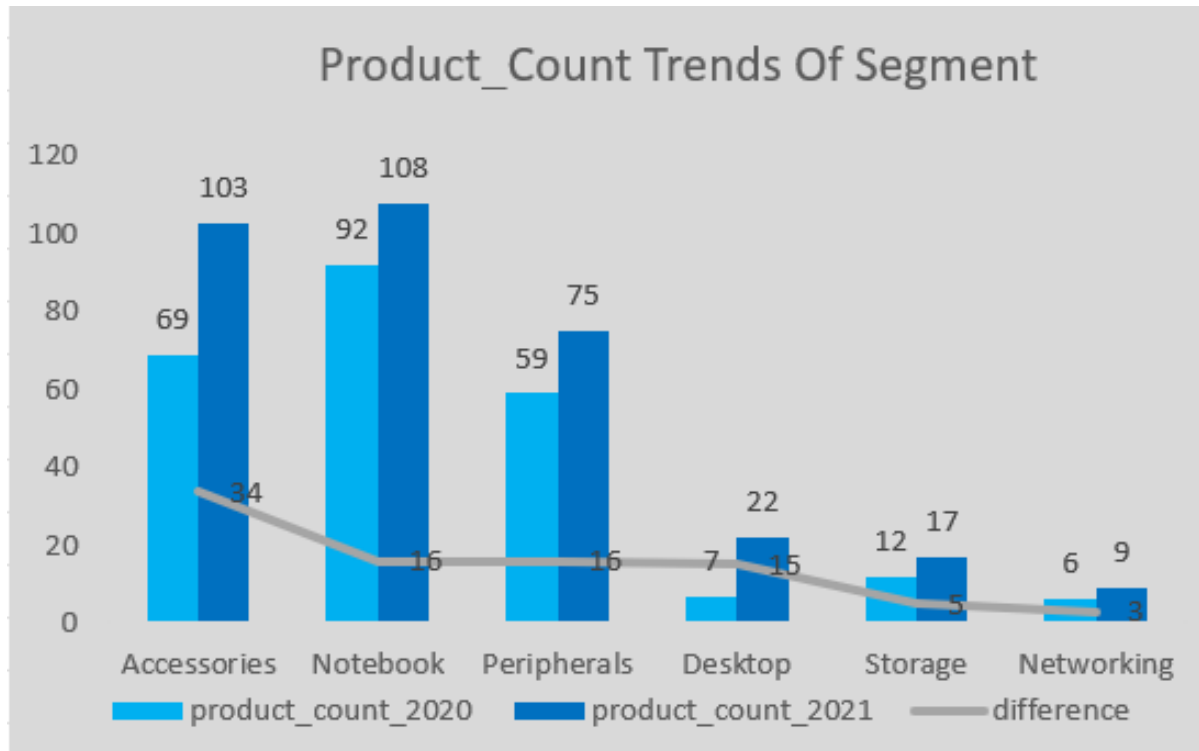
```
Question-4

with info_ as (
select
    segment,
    count(distinct case when fiscal_year = 2020 then fact_sales_monthly.product_code
end) as product_count_2020,
    count(distinct case when fiscal_year = 2021 then fact_sales_monthly.product_code
end) as product_count_2021
from
    fact_sales_monthly inner join dim_product
    on (fact_sales_monthly.product_code = dim_product.product_code)
group by
    segment )

select
    segment,
    product_count_2020,
    product_count_2021,
    (product_count_2021 - product_count_2020) as difference
from
    info_ order by difference desc ;
```

	segment	product_count_2020	product_count_2021	difference
▶	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3

VISUALS- 4



Insights

- The "Accessories" segment saw a significant increase in product counts, with 34 more products in 2021 compared to 2020.
- These insights suggest a general trend of product portfolio growth, particularly in the "Accessories" and "Notebook" segments, which can indicate a strategy to cater to a broader range of customer preferences.
- The growth in product variety may enhance market competitiveness and offer customers more choices.

AD_HOC-REQUEST:5

QUESTION:

Get the products that have the highest and lowest manufacturing costs.

```
Question-5

SELECT
    fc.product_code, product, CONCAT(manufacturing_cost) AS manufacturing_cost
FROM
    fact_manufacturing_cost as fc
JOIN dim_product as dp
ON fc.product_code = dp.product_code
WHERE
    fc.manufacturing_cost = (SELECT max(manufacturing_cost) FROM
fact_manufacturing_cost) OR
    fc.manufacturing_cost = (SELECT min(manufacturing_cost) FROM
fact_manufacturing_cost)
ORDER BY manufacturing_cost DESC;
```

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

Insights

- "AQ HOME Allin1 Gen 2" has a relatively higher manufacturing cost of 240.5364.
- In contrast, "AQ Master wired x1 Ms" has a significantly lower manufacturing cost of 0.892

AD_HOC-REQUEST:6

QUESTION:

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.

```
Question-6

select
    dim_customer.customer_code,
    customer,
    round(avg(pre_invoice_discount_pct)*100,2) as average_discount_percentage
from
    dim_customer right join fact_pre_invoice_deductions
    on (dim_customer.customer_code = fact_pre_invoice_deductions.customer_code)
where
    fiscal_year = 2021 and market = 'India'
group by
    customer_code,
    customer
order by average_discount_percentage DESC
LIMIT 5;
```

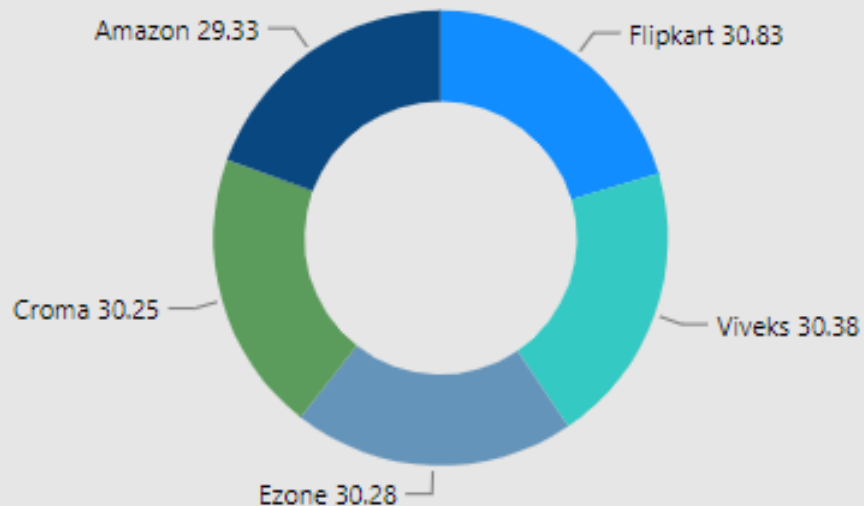
	customer_code	customer	average_discount_percentage
▶	90002009	Flipkart	30.83
	90002006	Viveks	30.38
	90002003	Ezone	30.28
	90002002	Croma	30.25
	90002016	Amazon	29.33

VISUALS- 6

Insights

- "Flipkart" has the highest average pre-invoice discount percentage at 30.83%.
- "Amazon" has a comparatively lower average discount at 29.33%.
- These insights indicate varying discount strategies among customers, with "Flipkart" and "Viveks" offering the highest average discounts, potentially attracting cost-conscious shoppers.
- "Amazon" provides relatively lower discounts, suggesting a different pricing strategy or a customer base less sensitive to discounts.
- These insights can be valuable for adjusting discount strategies and understanding customer preferences

customer vs avg_dis_percentage



AD_HOC-REQUEST:7

QUESTION:

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month.

Question-7

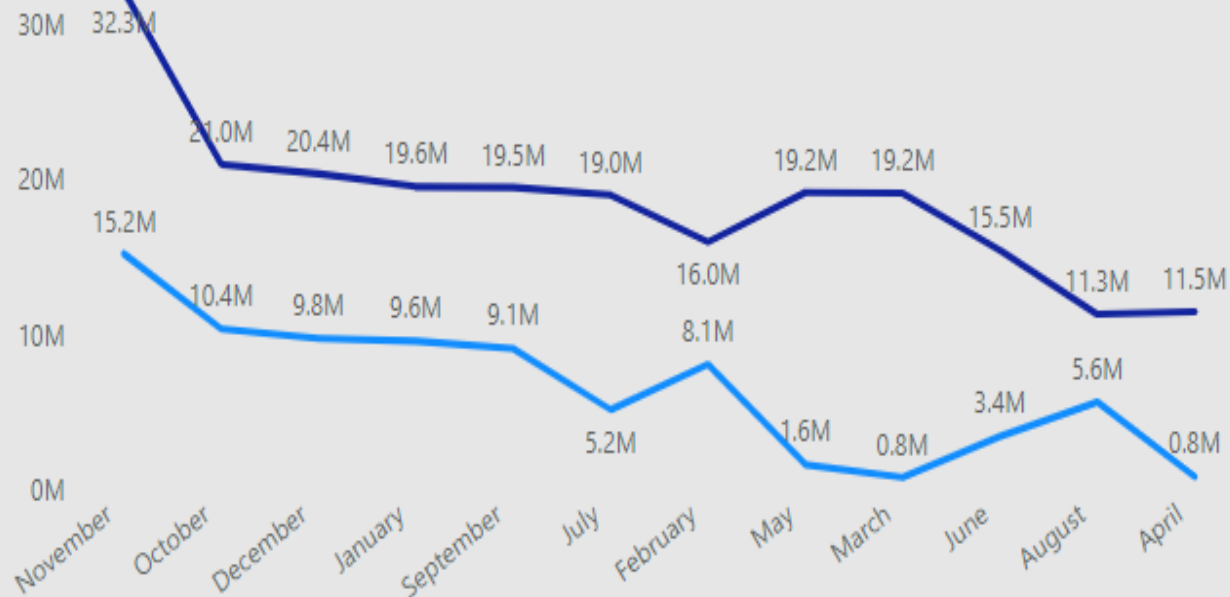
```
select
    monthname(date) as month_,
    fact_sales_monthly.fiscal_year,
    concat(round(sum(gross_price * sold_quantity) / 1000000, 2), ' M') as
gross_sales_amount
from
    fact_sales_monthly
    left join fact_gross_price on (fact_sales_monthly.product_code =
fact_gross_price.product_code)
    left join dim_customer on (fact_sales_monthly.customer_code =
dim_customer.customer_code)
where
    customer = 'Atliq Exclusive'
group by
    monthname(date),
    fact_sales_monthly.fiscal_year
order by
    fact_sales_monthly.fiscal_year ;
```

	month_	fiscal_year	gross_sales_amount
▶	September	2020	9.09 M
	October	2020	10.38 M
	November	2020	15.23 M
	December	2020	9.76 M
	January	2020	9.58 M
	February	2020	8.08 M
	March	2020	0.77 M
	April	2020	0.80 M
	May	2020	1.59 M
	June	2020	3.43 M
	July	2020	5.15 M
	August	2020	5.64 M
	September	2021	19.53 M
	October	2021	21.02 M
	November	2021	32.25 M
	December	2021	20.41 M
	January	2021	19.57 M
	February	2021	15.99 M
	March	2021	19.15 M
	April	2021	11.48 M
	May	2021	19.20 M
	June	2021	15.46 M
	July	2021	19.04 M
	August	2021	11.32 M

VISUALS- 7

Gross_Sales_amt by month_ and fiscal_year

fiscal_year ● 2020 ● 2021



Insights

- November 2021 had the highest gross sales amount, reaching \$32,247,289.79.
- In contrast, the fiscal year 2021 started with lower sales in September but still had a significant peak in November.
- There is a notable seasonality in sales, with November being a consistently strong month.
- The months of March and April in fiscal year 2020 had relatively low sales, which improved in fiscal year 2021.
- These insights can guide strategic decisions, such as focusing marketing efforts and inventory planning around the peak sales months and addressing potential challenges during lower sales months.

AD_HOC-REQUEST:8

QUESTION:

In which quarter of 2020, got the maximum total_sold_quantity?

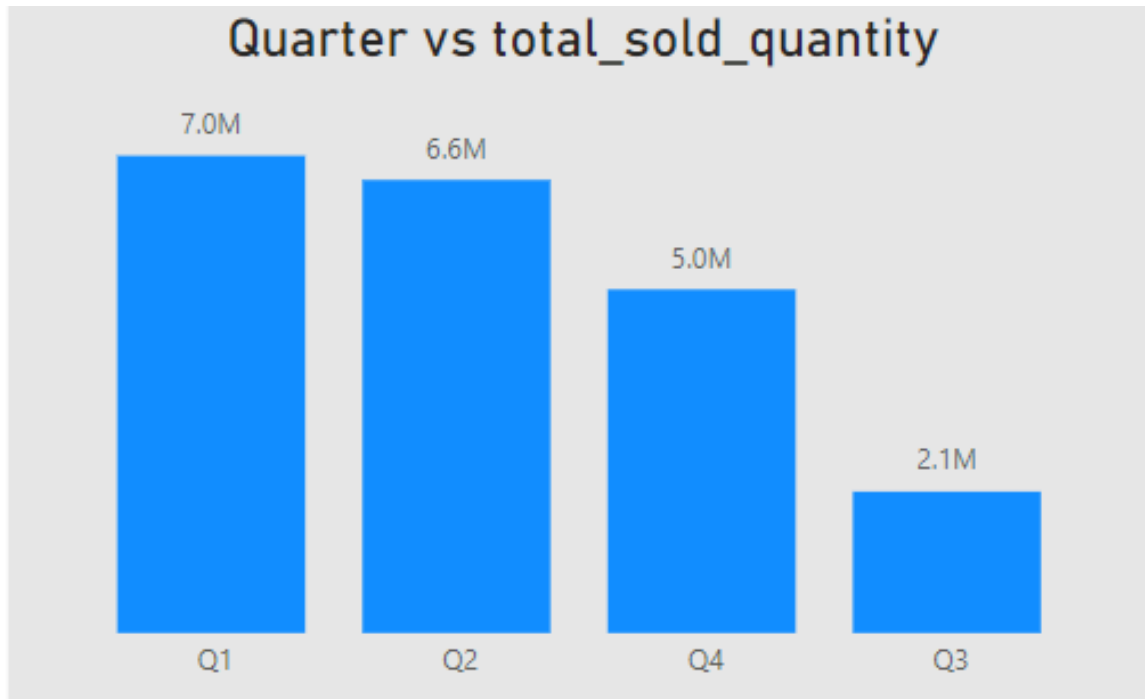
```
Question- 8

WITH quarter AS (
    SELECT sold_quantity,
           CASE
               WHEN MONTH(date) BETWEEN 09 AND 11 THEN "Q1"
               WHEN MONTH(date) IN (12, 01, 02) THEN "Q2"
               WHEN MONTH(date) BETWEEN 03 AND 05 THEN "Q3"
               WHEN MONTH(date) BETWEEN 06 AND 08 THEN "Q4"
           END as Quarter
    FROM fact_sales_monthly
    WHERE fiscal_year = 2020)

SELECT Quarter, SUM(sold_quantity) AS total_sold_quantity FROM quarter
GROUP BY Quarter
ORDER BY total_sold_quantity DESC;
```

	Quarter	total_sold_quantity
▶	Q1	7005619
	Q2	6649642
	Q4	5042541
	Q3	2075087

VISUALS- 8



Insights

- The highest total sold quantity is in Q1, with 7,005,619 units.
- These insights highlight a seasonal variation in sales, with Q1 and Q2 being the strongest quarters and Q3 being the weakest.
- This information is valuable for planning inventory and marketing strategies to align with seasonal demand

AD_HOC-REQUEST:9

QUESTION:

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

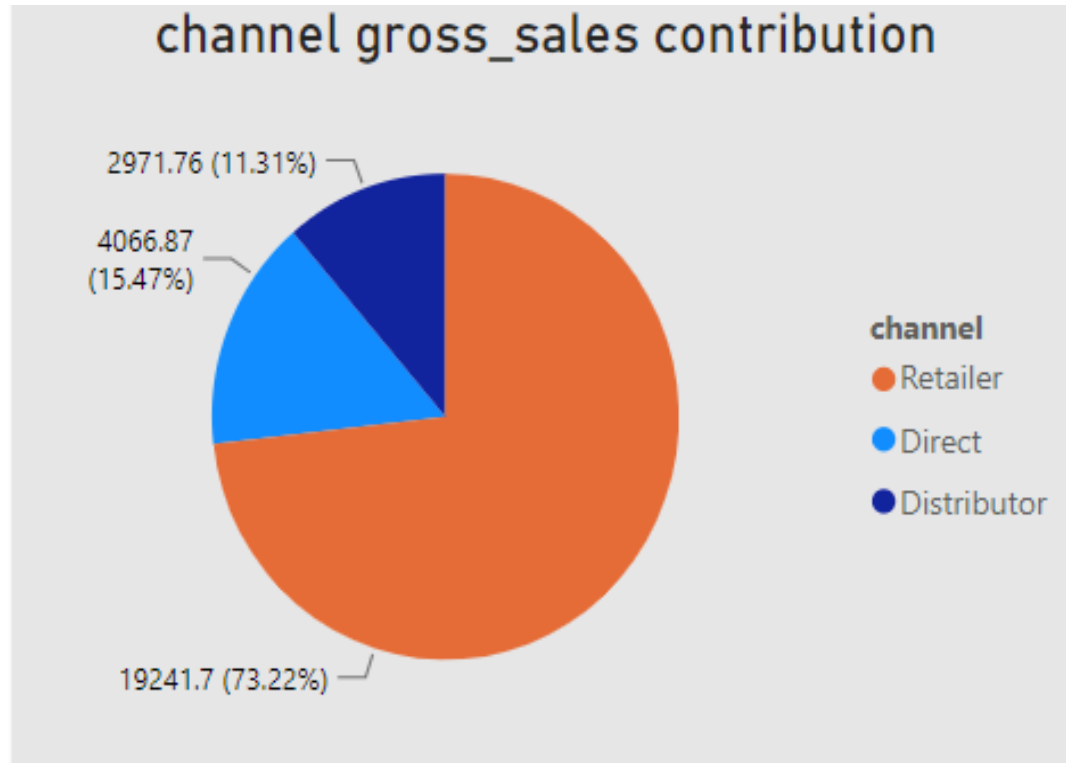
```
Question- 9

with x as (select
  channel,
  (round(sum(s.sold_quantity * g.gross_price) / 100000, 2)) as gross_sales_mln
from
  fact_sales_monthly s
  join fact_gross_price g on (s.product_code = g.product_code)
  join dim_customer c on (s.customer_code = c.customer_code)
where
  s.fiscal_year = 2021
group by
  c.channel)
select channel, gross_sales_mln,
round((gross_sales_mln/(select sum(gross_sales_mln) from x))*100,2)
as pct from x
order by gross_sales_mln desc;
```

	channel	gross_sales_mln	pct
▶	Retailer	19241.70	73.22
	Direct	4066.87	15.47
	Distributor	2971.76	11.31

VISUALS- 9

Insights



- The "Retailer" channel accounts for the majority of sales, contributing to 73.22% of gross sales.
- The "Direct" channel also plays a significant role, representing 15.47% of gross sales.
- The "Distributor" channel contributes 11.31% of gross sales. A significant focus on the "Retailer" channel suggests it is the primary revenue driver.
- Diversification and growth opportunities may be explored in the "Direct" and "Distributor" channels to further maximize sales.

AD_HOC-REQUEST: 10

QUESTION:

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

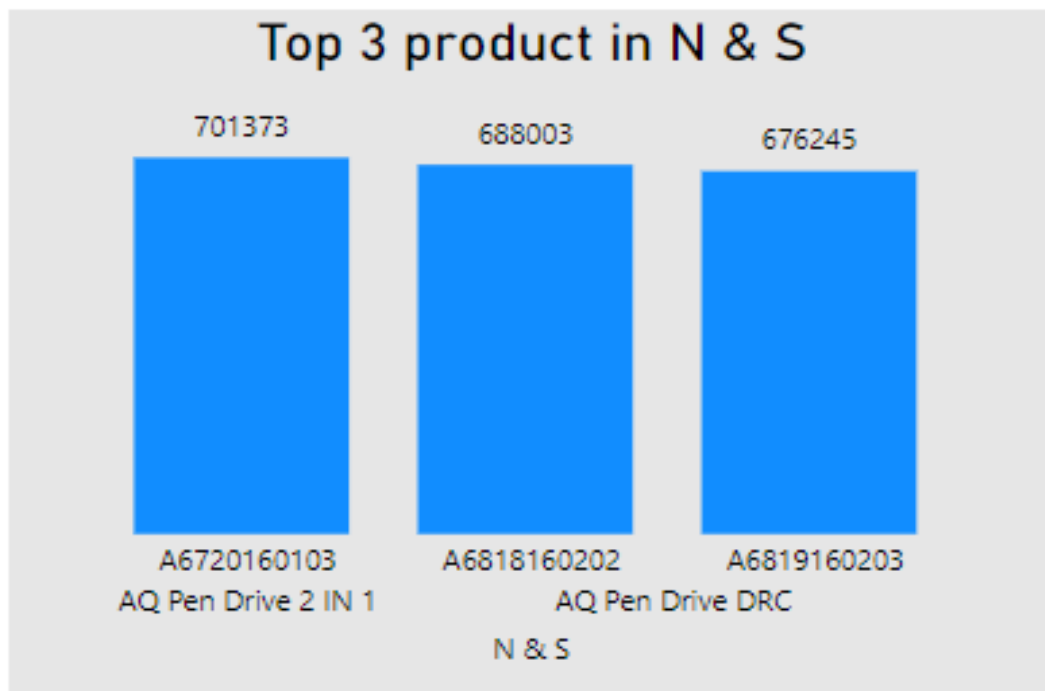
```
Question- 10

WITH x AS (
  SELECT
    P.division,
    S.product_code,
    P.product,
    SUM(S.sold_quantity) AS Total_sold_quantity,
    RANK() OVER(PARTITION BY P.division ORDER BY SUM(S.sold_quantity) DESC) AS
Rank_Order
  FROM
    dim_product P
  JOIN
    fact_sales_monthly S ON P.product_code = S.product_code
  WHERE
    S.fiscal_year = 2021
  GROUP BY
    P.division, S.product_code, P.product
)
SELECT *
FROM x
WHERE Rank_Order IN (1, 2, 3)
ORDER BY division, Rank_Order;
```

	division	product_code	product	Total_sold_quantity	Rank_Order
▶	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
	N & S	A6818160202	AQ Pen Drive DRC	688003	2
	N & S	A6819160203	AQ Pen Drive DRC	676245	3
	P & A	A2319150302	AQ Gamers Ms	428498	1
	P & A	A2520150501	AQ Maxima Ms	419865	2
	P & A	A2520150504	AQ Maxima Ms	419471	3
	PC	A4218110202	AQ Digit	17434	1
	PC	A4319110306	AQ Velocity	17280	2
	PC	A4218110208	AQ Digit	17275	3

VISUALS - 10

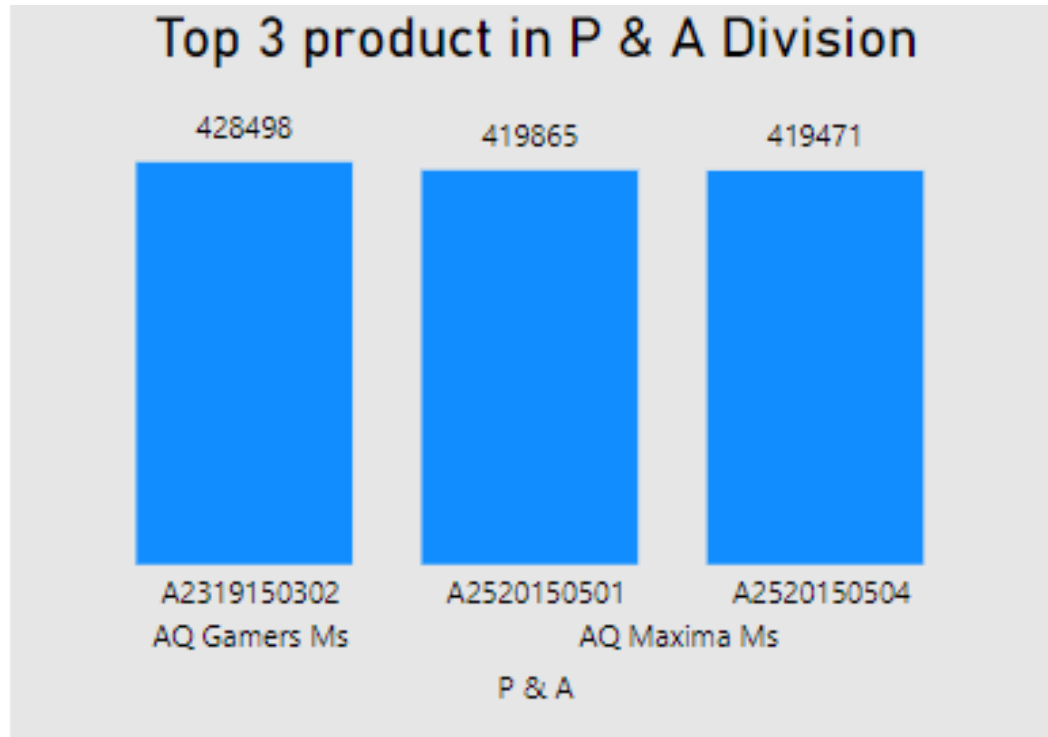
Insights



- The top three products: "AQ Pen Drive 2 IN 1," "AQ Pen Drive DRC," and another variant of "AQ Pen Drive DRC Strong consumer preference for pen drives and related items in this division.
- The division's leadership in this market segment, indicating growth potential.
- Efficient inventory management is crucial for sustained success

VISUALS - 10

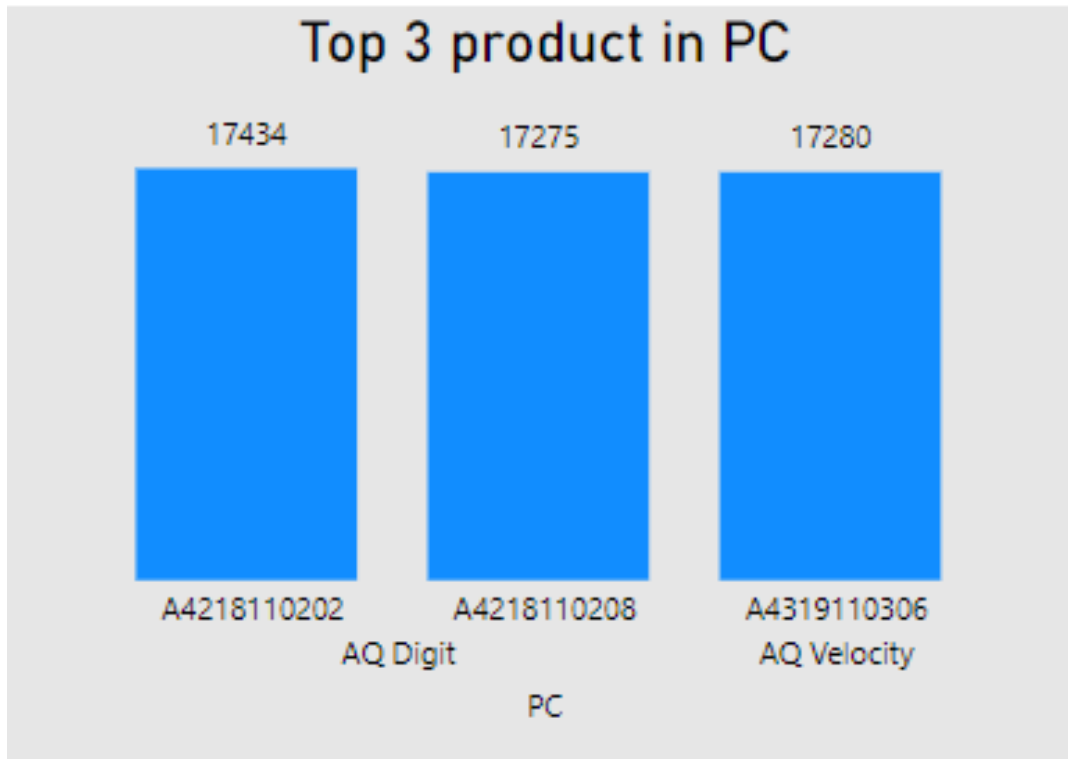
Insights



- The best-selling product is "AQ Gamers Ms. " followed by two variants of "AQ Maxima Ms."
- This division specializes in gaming-related products, appealing to a specific customer niche..
- Efficient inventory management is essential to meet customer demand.
- Continuously promote and brand the top products to sustain success

VISUALS - 10

Insights



- The best-selling product in the "PC" division is "AQ Digit," with two variants followed by "AQ velocity".
- The division specializes in PC-related products, indicating a focus on the computing niche.
- Efficient inventory management is essential to meet customer demand.
- Ongoing branding and marketing efforts are important to maintain success



Thank You

